

Steady State Analysis of Nonlinear Dielectrics Filled Microwave Passive Devices

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Abstract—This letter presents the computation of high order harmonic components of the time-harmonic field exciting waveguide structures partially filled with nonlinear dielectrics. The approach presented employs the Harmonic Balance method to perform a multi-harmonic Finite Element analysis, and the solution of the nonlinear system is handled either with a Picard or a relaxed iteration, depending on the strength of the nonlinear behavior. Since the nonlinear loop is particularly time demanding, a domain decomposition method is here proposed to speed-up the system solution, restricting the iteration loop to small sub-domain.

Index Terms—Harmonic Balance, Finite Elements, Domain Decomposition, nonlinear dielectrics, waveguide filter

I. INTRODUCTION

TODAY's microwave and millimeter wave passive devices design require, when high power densities are involved, the modeling of material nonlinear properties to accurately predict the steady state electromagnetic field. Among the available techniques, a time-stepping approach to the solution of Maxwell's equations in such cases results to be computationally prohibitive, the transient period having to vanish. An alternative time-harmonic approach, based on the finite element (FE) solution of a partial differential equation (PDE) of time-harmonic variable, provides the field quantities at a given frequency. Unfortunately, a single frequency is not sufficient to include all relevant phenomena, high order harmonics being generated from the interaction of the electromagnetic field with the nonlinear materials. Incorporating a larger set of harmonics in the FE formulation by using the Harmonic Balance Finite Element (HBFE) method [1], [2], also known as Multi-Harmonic Finite Element method [3]–[6], provides a fast and accurate solution to the problem as long as the harmonics required to affordably represent the field is not excessive. The nonlinearity of the PDE is tackled by an iterative solution of a spatially and spectrally coupled large-scale linear system updated at each iteration with a new linearization such that the computed solutions converge, within a prescribed tolerance, to the sought solution. An algebraic Domain Decomposition technique as proposed in [7] improves computational timings, especially when the region occupied by nonlinear materials is rather small.

In this letter, we focus on the analysis of microwave devices filled with dielectrics presenting nonlinear scalar relative permittivity. In section II, we present the Harmonic Balance Finite

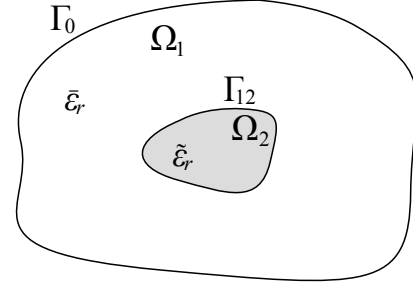


Fig. 1. Sketch of the domain $\Omega = \Omega_1 \cup \Omega_2$.

Element method formulation for the solution of the vector Helmholtz equation for the electric field, within a domain comprising nonlinear dielectrics. In section III, simulation results of a broadband E-plane bandpass filter filled with Kerr-type material are shown, and the performances of the algorithm are discussed.

II. FORMULATION

The formulation given in this section holds for the case of a generic electromagnetic problem defined on a closed domain as depicted in Fig. 1. The domain Ω , with boundary Γ_0 , can be either defined on $\mathbb{R}^2$ or $\mathbb{R}^3$. For the sake of simplicity, the examples presented in the next section are analyzed in $\mathbb{R}^2$, allowing for straightforward implementation. Ω is divided into a large part, Ω_1 , with linear scalar relative permittivity $\bar{\epsilon}_r(\mathbf{r})$ and a small part, Ω_2 , with a nonlinear relative permittivity $\tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}))$. The scalar permittivity in Ω can be expressed as

$$\epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \begin{cases} \bar{\epsilon}_r(\mathbf{r}), & \mathbf{r} \in \Omega_1 \\ \tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r} \in \Omega_2 \end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field pointed by $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator that describes the analytically known nonlinear behavior. The electric field satisfies, within Ω , the vector Helmholtz equation

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \right) - k_0^2 \epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \quad (2)$$

with suitable boundary conditions on Γ_0 . For the example shown in section III, Γ_0 is considered partly perfect electric conductor, partly a modal expansion of the field at the wave ports as developed in [8].

We proceed with the standard FE Galerkin framework: the domain Ω is discretized into a set of non overlapping finite

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elements and the electric field is approximated as a linear combination of vector basis functions α_n

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^N E_n \alpha_n(\mathbf{r}) \quad (3)$$

Then by testing (2) over Ω with the weighting functions $\mathbf{W}_m$, chosen the same as the shape functions α_n , we obtain the following weak form:

$$\begin{aligned} & \int_{\Omega} \nabla \times \mathbf{W}_m(\mathbf{r}) \cdot \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \, d\Omega \\ & - k_0^2 \int_{\Omega} \mathbf{W}_m(\mathbf{r}) \cdot \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) \, d\Omega \\ & + \oint_{\Gamma_0} \mathbf{W}_m(\mathbf{r}) \cdot \hat{n} \times \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \, d\Gamma = 0 \end{aligned} \quad (4)$$

We introduce the multi-harmonic dependence of the HBF method, by approximating the electric field as a truncated Fourier series [4]

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{p=1}^P \mathbf{E}_p^{(s)}(\mathbf{r}) \sin(p\omega t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(p\omega t) \quad (5)$$

where ω is the fundamental angular frequency. The Fourier expansion does not comprehend any dc-term as a static field does not represent a solution to (2) for $k_0 > 0$. $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the finite element coefficients E_n , and they must individually satisfy (2).

The scalar relative permittivity can also be approximated as a truncated Fourier series

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}, t) = \varepsilon_{r_0} + \sum_{k=1}^K \varepsilon_{r_k}^{(s)} \sin(k\omega t) + \varepsilon_{r_k}^{(c)} \cos(k\omega t) \quad (6)$$

and the coefficients of the expansion are given by

$$\varepsilon_{r_0} = \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \, dt \quad (7)$$

$$\varepsilon_{r_k}^{(s)} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \sin(k\omega t) \, dt \quad (8)$$

$$\varepsilon_{r_k}^{(c)} = \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \cos(k\omega t) \, dt \quad (9)$$

with $k = 1 \dots K$. The approximation orders P and K of, respectively, the electric field and the scalar permittivity, depend on the specific device analyzed and on the intensity of the incident field. These can be set by choosing an appropriate convergence criterion of the power associated to the harmonics of lower order as the number of multi-harmonic coefficients is increased [9].

We proceed substituting $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (4), then testing the equation with $\sin(m\omega t)$ and $\cos(m\omega t)$, $m = 1 \dots M$, and integrating over the time period of the fundamental $[0, \frac{2\pi}{\omega}]$

gives a large-scale linear system of the form:

$$\begin{bmatrix} a_{11}^{(1s1s)} & a_{11}^{(1s1c)} & \dots & a_{1N}^{(1sMc)} \\ a_{11}^{(1c1s)} & a_{11}^{(1c1c)} & \dots & a_{1N}^{(1cMc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1}^{(Mc1s)} & a_{N1}^{(Mc1c)} & \dots & a_{NN}^{(McMc)} \end{bmatrix} \begin{bmatrix} E_1^{(1s)} \\ E_1^{(1c)} \\ \vdots \\ E_N^{(Mc)} \end{bmatrix} = \begin{bmatrix} b_1^{(1s)} \\ b_1^{(1c)} \\ \vdots \\ b_N^{(Mc)} \end{bmatrix} \quad (10)$$

The product that occurs in the second integral of (4), between the approximated field and permittivity, induces coupling between the harmonic components. However, as the PDE we seek to solve is free of first order time derivative, there is no coupling between $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$. Thus, we may retain either the sine or the cosine harmonic function to excite the electromagnetic device and to express the multi-harmonic behavior of the field, halving in such a way the system dimensions. The order of the resulting HBF system is NP . In practical analyses, this number can become considerable, and as introduced previously, the solutions have to be computed iteratively, building at each step a new linearized system, until convergence of the solution is achieved.

The chosen nonlinear solution is based on a relaxed iteration, with $\gamma \in [0, 1]$ the relaxation coefficient, and $\gamma = 1$ constitutes the particular case of Picard iteration. The permittivity is assumed initially to be the one in the absence of electric field:

$$[A_{([E]^0)}][E]^1 = [b_{([E]^0)}] \quad (11)$$

where $[E]^0$ is a null vector of unknowns. The solution of the first linear system of (11) provides expansion coefficients that will be used to compute a new value of the permittivity:

$$[A_{(\gamma[E]^{q-1} + (1-\gamma)[E]^{q-2})}][E]^q = [b_{(\gamma[E]^{q-1} + (1-\gamma)[E]^{q-2})}] \quad (12)$$

with $q \geq 2$. The process is repeated until the relative error between the updated solution and the previous one, in the sense of the euclidean norm, is less than a prescribed value τ :

$$\frac{\|[E]^q - [E]^{q-1}\|}{\|[E]^q\|} \leq \tau \quad (13)$$

For the example of section III, $\tau = 10^{-6}$ have been chosen. Highly nonlinear materials and high intensity impressed fields may hamper the convergence of the nonlinear iteration, due to the oscillation of the solutions around the expected values. Lowering the value of γ has the effect to damp that oscillation, hence rendering the iteration more robust but inevitably reducing the convergence rate [10].

To enhance the computational efficiency, a substructuring domain decomposition approach is used [7], [11]–[13]. The multiharmonic unknown vector $[E]$ defined by the HBF method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2 , respectively. Unknowns belonging to the boundaries are placed in a third vector $[E^c]$. The HBF system can then

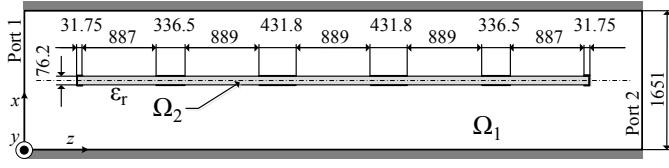


Fig. 2. Cross section of a broadband E-plane bandpass filter realized in an empty WR6 rectangular waveguide [14]. Measures are given in μm .

be recast in the form:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M_{([E]^{q-1})}^{(2)}] & [P_{([E]^{q-1})}^{(2)}] \\ [Q^{(1)}] & [Q_{([E]^{q-1})}^{(2)}] & [C_{([E]^{q-1})}] \end{bmatrix} \begin{bmatrix} [E^{(1)}]^q \\ [E^{(2)}]^q \\ [E^{(c)}]^q \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b_{([E]^{q-1})}^{(2)}] \\ [b_{([E]^{q-1})}^{(c)}] \end{bmatrix} \quad (14)$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$ fully internal to Ω_1 , hence with null harmonic coupling coefficients. $[M^{(2)}]$ contains coefficients for the nonlinear sub-domain Ω_2 . $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between the interior unknowns and the ones on the boundaries which are collected in $[E^{(c)}]$. By using the Schur complement $[S_{([E]^{q-1})}]$, constructed with a Gauss elimination-type strategy, the boundary unknowns $[E^{(c)}]^q$ of (14) are computed by solving the reduced system

$$[S_{([E]^{q-1})}][E^{(c)}]^q = [d]^{q-1} \quad (15)$$

where

$$[S_{([E]^{q-1})}] = [C_{([E]^{q-1})}] - [Q^{(1)}][M^{(1)}]^{-1}[P^{(1)}] - [Q_{([E]^{q-1})}^{(2)}][M_{([E]^{q-1})}^{(2)}]^{-1}[P_{([E]^{q-1})}^{(2)}] \quad (16)$$

and

$$[d]^k = [b_{([E]^{q-1})}^{(c)}] - [Q^{(1)}][M^{(1)}]^{-1}[b^{(1)}] - [Q_{([E]^{q-1})}^{(2)}][M_{([E]^{q-1})}^{(2)}]^{-1}[b_{([E]^{q-1})}^{(2)}] \quad (17)$$

The difference operators in (16) and (17) are to be intended as an iterative assembling of local matrices in a global one [12]. Once $[E^{(c)}]^q$ is known, local solutions are given by

$$[E^{(1)}]^q = [M^{(1)}]^{-1} \left([b^{(1)}] - [P^{(1)}][E^{(c)}]^q \right) \quad (18)$$

$$[E^{(2)}]^q = [M_{([E]^{q-1})}^{(2)}]^{-1} \left([b_{([E]^{q-1})}^{(2)}] - [P_{([E]^{q-1})}^{(2)}][E^{(c)}]^q \right) \quad (19)$$

In order to solve the HBFE system and compute the steady state field, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 and its boundary are to be updated. The efficiency of this DD technique relies on the fragmentation of the matrices and the subsequent computation of partial solutions. When the number of coefficients related to nonlinear media is small, as usually happens for microwave devices where the domain of nonlinear media has a reduced extension, noticeable improvements can be achieved.

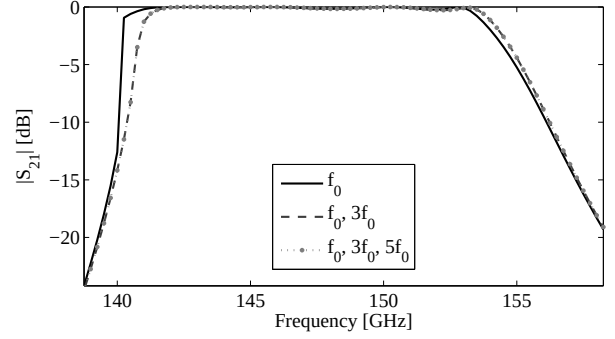


Fig. 3. Behavior of the spectral response as the number of harmonics retained for the HBFE increases. $E_i = 10 \text{ kV/m}$.

III. NUMERICAL RESULTS

In this section, the analysis of a millimeter wave broadband filter [14] is performed and the computational results are shown. The filter is realized within a standard WR6 waveguide by properly designed metalizations on a $6040.5 \mu\text{m}$ long dielectric slab mounted along the E-plane at the middle of the width range. A sketch of the H-plane cross section of the filter, plane of the performed 2D HBFE analysis, is shown in Fig. 2. First order nodal elements are used and the domain discretization is such that 2470 degrees of freedom are obtained, 42 belonging to the nonlinear subdomain Ω_2 , 456 to the boundaries and the remaining 1972 to the linear subdomain Ω_1 . The metalizations are considered perfectly conducting, thus imposing the appropriate Dirichlet conditions. The continuity of the field at the ports is imposed through a modal expansion which, for the geometry of the problem, consists of TE_{n0} , $n \in \mathbb{Z}$, modes where $n \leq 10$ have been chosen. The field incident at Port 1 is of only a TE_{10} mode.

The dielectric, enclosed in Ω_2 , presents a Kerr-type nonlinearity such that [15]

$$\bar{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \alpha_2 |\mathbf{E}(\mathbf{r})|^2, \quad \mathbf{r} \in \Omega_2 \quad (20)$$

where $\bar{\epsilon}_r(\mathbf{r}) = 2.1$ and α_2 is chosen of $1.625 \cdot 10^{-10} \text{ m}^2/\text{V}^2$ according to [16]. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus the even ones have been neglected in the truncated Fourier expansion. Furthermore, as we choose a sinusoidal excitation, only $\sin(m\omega t)$ related coefficients of the electric field are retained.

The analysis of the effects of the nonlinearity on the spectrum is performed with an increasing intensity of the field. $E_i = |\mathbf{E}(\mathbf{r})|_{\text{max}}$ assumes the values of 5 kV/m, 10 kV/m and 15 kV/m corresponding to 18 mW, 72 mW and 162 mW of incident power, respectively, which are common power values encountered in D-band applications [17]. Results for the guided TE_{10} are shown in Fig. 6, building up the HBFE system up to the 7th harmonic ($7f_0$). There is an evident shift the bandwidth range to lower frequencies as higher power densities are involved.

The harmonic power all over the device and its convergence as the number of harmonics is increased is shown in Fig., which emphasizes that as we increase the number of harmonics, the lower order ones, especially the fundamental, present

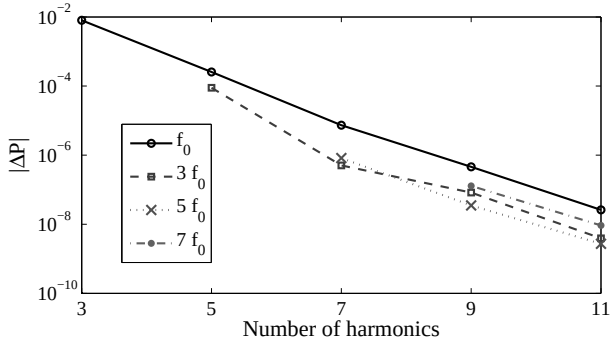


Fig. 4. Converging behavior of the power associated to the harmonics as the number of the harmonics employed increases when $P_{in} = 200$ mW at 150 GHz.

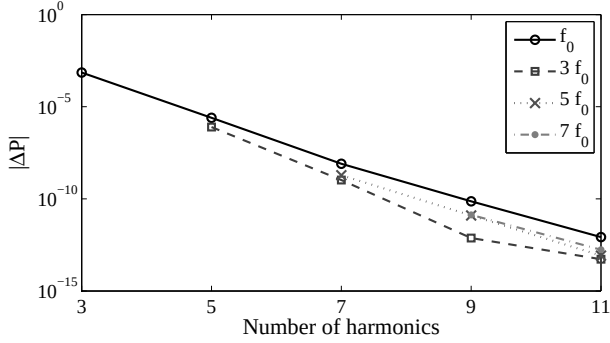


Fig. 5. Converging behavior of the power associated to the harmonics as the number of the harmonics employed increases when $P_{in} = 72$ mW at 150 GHz.

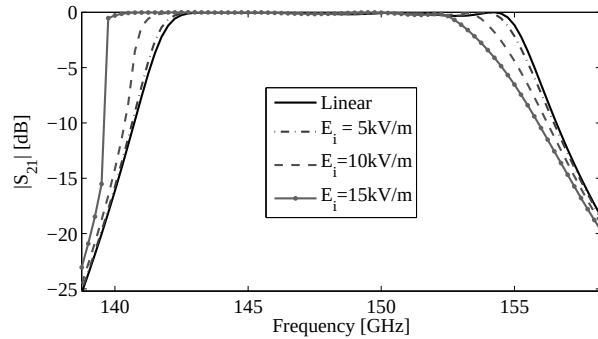


Fig. 6. S_{21} amplitude response of the filter for the TE_{10} guided mode computed including up to the 7th harmonic.

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a slight variation. This allows for limiting the order of the HBFE system.

The effect of the number of harmonics on the spectrum is shown in Fig. 3

- The Domain decomposition technique induces noticeable improvements in the simulation timings, especially for high number of harmonics

As we can show

IV. CONCLUSION

The conclusion goes here.